

KEYbooks (SmartERP)

Architecture, Feature Breakdown & Technical Documentation

AUTHOR & ARCHITECT

Antra Sharma

CORE PARADIGM

Zero-Mouse Accounting

PRIMARY STACK

Next.js 15 • Express 5 •
PostgreSQL

DOCUMENTATION DATE

Academic Year 2026

1. Executive Summary

KEYbooks (SmartERP) is a keyboard-driven, full-stack financial enterprise resource planning (ERP) system designed to bridge the gap between traditional desktop accounting software (such as **Tally Prime**) and modern cloud-native web applications (such as **Zoho Books** and **QuickBooks**).

Built as an academic and internship capstone project by **Antra Sharma**, KEYbooks delivers a zero-mouse bookkeeping experience directly in the browser. It combines rigorous **double-entry bookkeeping validation**, **FIFO / Weighted Average Cost (WAC) inventory valuation**, **real-time financial statement generation** (Trial Balance, Profit & Loss, Balance Sheet), and **multi-user mutex write locks** to simulate an enterprise-grade ERP architecture.

2. Core Purpose & Problem Statement

2.1 The Problem with Existing Financial Softwares

Traditional accounting workflows present two opposing extremes in the market today:

1. Desktop Legacy Systems (e.g., Tally Prime)

Strengths: Lightning-fast keyboard-only entry; accounting clerks can post hundreds of vouchers per day without touching a mouse.

Weaknesses: Monolithic local database files, challenging remote accessibility, lack of cloud collaboration, rigid desktop-bound installation, and dated interfaces.

2. Modern Cloud SaaS (e.g., QuickBooks, Zoho Books)

Strengths: Accessible anywhere via web browser, responsive UI, cloud backups, multi-user role management.

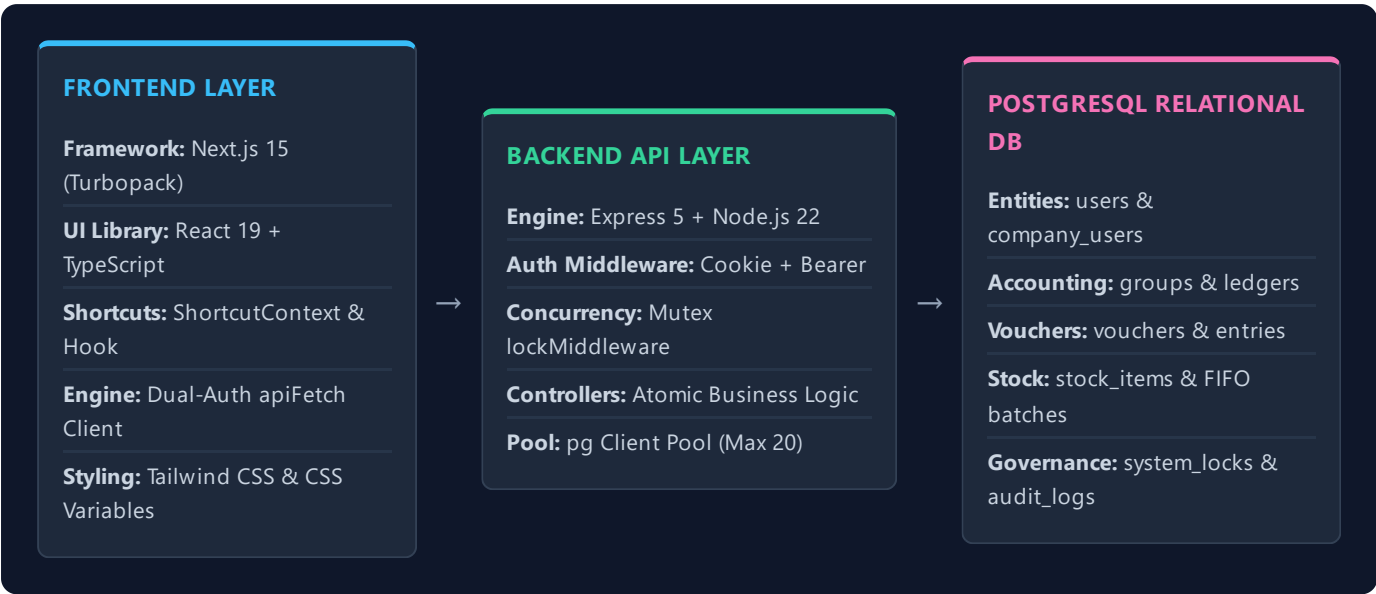
Weaknesses: Mouse-heavy point-and-click forms that slow down skilled data-entry clerks; heavy form validations that interrupt batch entry rhythms; delayed report updates relying on background caching.

2.2 KEYbooks' Unified Solution

The Hybrid Philosophy of KEYbooks:

KEYbooks harmonizes these worlds by providing **native browser accessibility** paired with a **strict keyboard-first design**: function keys (**F1**–**F9**), **Alt** hotkeys, **Ctrl+K** omnisearch, and sequential field progression via **Enter** / **Arrow** keys. Mathematical integrity is guaranteed at the database level via atomic transactions where $\sum \text{Debits} = \sum \text{Credits}$.

3. System Architecture & Tech Stack



3.1 Technology Stack Details

Layer	Technology	Version & Details	Primary Responsibility
Frontend Framework	Next.js	15.x (App Router, Turbopack)	Server-side rendering, client hydration, route layouts
Client Core	React	19.x, TypeScript	State management, component tree, modal lifecycle
Styling & Design	Tailwind CSS	4.x + Vanilla CSS Variables	Dark/Light mode tokens, micro-animations, responsive layout
Icons & Animation	Lucide React + Lenis	Lucide icons, Lenis smooth scroll	Visual indicators, landing page physics
Backend Framework	Express.js	5.x, Node.js 18+	REST API endpoints, routing, middleware orchestration
Database	PostgreSQL	14+ (pg connection pool)	Relational integrity, ACID transactions, check constraints
Security & Auth	JWT, Bcrypt, Helmet	Dual auth (Cookie + Bearer), 12 salt rounds	Session tokens, password hashing, security headers
Mailing Service	Nodemailer	SMTP integration	Account verification, password reset links

4. Key Functional Modules

4.1 Multi-Company Management

Single user account handles up to **5 distinct legal entities** or subsidiaries.

- Distinct fiscal years (e.g. 2026-04-01 to 2027-03-31)
- Localized currencies (₹ , \$, € , £) and GSTIN handling
- Instant workspace switching via numeric keys [1] – [5]

4.2 Chart of Accounts (COA)

Strict 4-pillar double-entry hierarchy governing ledger accounts:

- **Assets & Expenses:** Normal Debit balance
- **Liabilities & Income:** Normal Credit balance
- Unlimited nested sub-groups with opening balances
- ON DELETE RESTRICT protection prevents orphan transactions

4.3 Double-Entry Voucher Engine

Supports **Payment, Receipt, Journal, Sales, and Purchase**.

$$\Sigma \text{ Debit Amounts} - \Sigma \text{ Credit Amounts} = 0$$

- Atomic database transactions guarantee rollback upon imbalance
- Opposite ledger mapping auto-populates contra records

4.4 Advanced Inventory (FIFO & WAC)

Tracks material valuation and stock balances in real time:

- **FIFO:** Matches oldest inward purchase batches against outward vouchers
- **WAC:** Continuously recalculates weighted average unit cost
- Dynamic reorder-level alerts prevent inventory stockouts

4.5 Live Dynamic Statements

Zero cached summaries or delayed nightly jobs:

- **Trial Balance:** Live ledger net debit and credit reconciliation
- **Profit & Loss:** Real-time operational revenues vs direct/indirect costs
- **Balance Sheet:** Cumulative equity, liabilities, and assets
- **Day Book & Cash Flow:** Chronological audit registers

4.6 Concurrency & Mutex Locks

Prevents write conflicts across multi-tenant sessions:

- **system_locks:** Mutex write lock on fiscal settings and master data
- Automatic lock expiry and release upon save
- **audit_logs:** Complete immutable audit trail with JSON snapshots

5. Keyboard Navigation Architecture

KEYbooks replicates the ergonomic efficiency of desktop bookkeeping via a unified keyboard listener pattern implemented in `useKeyboardShortcuts.ts` and `ShortcutContext.tsx`:

Hotkey	Action Target	Behavior & Purpose
F1	Help / Shortcuts Modal	Opens full cheat-sheet overlay displaying all available shortcuts
F2	Fiscal Period	Quick-selects active Financial Year start and end dates
F3	Company Info	Opens company information drawer and entity properties
F8	Sales Voucher	Directly routes to Sales Invoice / Sales Voucher creation
F9	Purchase Voucher	Directly routes to Purchase Voucher / Inward Stock registration
Ctrl + K / Cmd + K	Command Palette	Global search across screens, vouchers, ledgers, and reports
Alt + C	Calculator Overlay	Toggles floating calculator with instant memory recall into active fields
Alt + L	Ledgers Master	Quick jump to Chart of Accounts and Ledger creation
Alt + G	Groups Master	Quick jump to Account Group hierarchies
Alt + S	Stock Catalog	Quick jump to Stock Item Catalog & Godown registers
Alt + V	Voucher Gateway	Quick jump to general voucher entry portal
Esc	Back / Gateway	Closes active modals; from a master page, navigates up to Dashboard
Enter / ↓ Arrow	Next Input Field	Advances focus to the subsequent form field without mouse clicking
Ctrl + Enter	Submit / Post	Commits and saves the current form or voucher immediately

6. Detailed Comparison: KEYbooks vs. Existing Software

Capability / Dimension	KEYbooks (SmartERP)	Tally Prime	QuickBooks Online	Zoho Books
Deployment & Platform	Modern Web (Next.js + Express)	Windows Desktop Executable	Cloud SaaS	Cloud SaaS
Keyboard Ergonomics	First-class native web keys (<code>F1</code> , <code>F9</code> , <code>Alt</code>)	Gold standard desktop keyboard	Limited mouse-centric UI	Limited mouse-centric UI
Double-Entry Integrity	Strict $\Delta = 0$ atomic check	Strict balance check	Auto-balanced hidden journals	Auto-balanced hidden journals
Stock Valuation Methods	Real-time FIFO & WAC selectable	Multi-method	FIFO only (Advanced tiers)	FIFO only
Concurrency Locking	Explicit Mutex Locks (<code>system_locks</code>)	File-level locking	Optimistic row locking	Optimistic locking
System Inspectability	Live "Architect Specs" in UI (SQL & schema)	Proprietary TDL (closed engine)	Proprietary closed API	Proprietary closed API
Authentication Flow	Dual: HTTP-only Cookie + Bearer Token	Windows user / Tally Vault	OAuth 2.0 / SSO	OAuth 2.0 / SSO
Database Engine	Standard Relational PostgreSQL	Proprietary Flat-File DB (.900)	Proprietary Cloud DB	Proprietary Cloud DB
Installation Overhead	Zero-install web client	Local setup + license server	Zero-install web client	Zero-install web client

7. Database Entity Relationship Architecture

KEYbooks enforces relational integrity across 12 normalized relational PostgreSQL tables:

users	PK: id	companies	PK: id	groups	PK: id
id	UUID	id	UUID	id	UUID
email	VARCHAR UNIQUE	user_id	UUID → users	company_id	UUID → companies
password_hash	TEXT (Bcrypt)	name	VARCHAR	name	VARCHAR
is_verified	BOOLEAN	gst_number	VARCHAR(15)	type	asset liab inc exp
role	VARCHAR (user/admin)	currency	VARCHAR (₹/\$/€/£)	parent_id	UUID → groups
ledgers	PK: id	vouchers	PK: id	voucher_entries	PK: id
id	UUID	id	UUID	id	UUID
company_id	UUID → companies	company_id	UUID → companies	voucher_id	UUID → vouchers
group_id	UUID → groups	voucher_number	VARCHAR	ledger_id	UUID → ledgers
name	VARCHAR	voucher_type	pmt rcpt jn sal pur	type	debit credit
opening_balance	NUMERIC(15,2)	date	DATE	amount	NUMERIC(15,2)

8. Key Technical Innovations in KEYbooks



1. Dual-Layer Resilient Auth

Supports HTTP-only cookies and Authorization: Bearer headers simultaneously. Guarantees cross-origin resilience and security against XSS session hijacking.



2. Transparent Architect Specs

Embedded telemetry tabs allow auditors, professors, and engineers to inspect live parameterized SQL queries, execution latency, and JSON payloads directly inside reports.



3. Desktop Hotkey Emulation

Custom event-driven keyboard listeners intercept browser default actions (**F1** , **F3** , **Alt**) to enable 100% mouse-free bookkeeping directly in the browser.



4. Enterprise Concurrency Mutex

Active table-level and document-level locking via PostgreSQL `system_locks` ensures multi-user setups avoid race conditions during concurrent company updates.